12311410 陈宇轩. Stochastic Processes HW 4.

1. <u>Proof</u>: $\Phi_X(u) = \mathbb{E}[e^{iuX}] = \mathbb{E}[e^{iu(X_1+X_2)}] = \mathbb{E}[e^{iuX_1} \cdot e^{iuX_2}]$

$= \mathbb{E}[e^{iuX_1}] \cdot \mathbb{E}[e^{iuX_2}]$

$X_1 \perp X_2$

$\Rightarrow e^{iuX_1} \perp e^{iuX_2}$. $= \exp(ia^T u - \frac{1}{2} u^T \Sigma u) \cdot \exp\left(\int_{\mathbb{R}^d} \lambda(e^{iu^T y} - 1)\mu(dy)\right)$. $\square$.

2. <u>Proof</u>: WTS: Every Lévy measure is σ-finite on $\mathbb{R}^d \setminus \{0\}$.

$\Leftrightarrow \exists \{A_k\}_{k=1}^{\infty}$, $\mathbb{R}^d \setminus \{0\} = \bigcup_{k=1}^{\infty} A_k$, with $\nu(A_k) < +\infty$, $\forall k$.

For Lévy measure ν: $\int_{\mathbb{R}^d \setminus \{0\}} \min\{|y|^2, 1\} \nu(dy) < +\infty$.

$$\boxed{\int_{A_k} \min\{|y|^2, 1\} \nu(dy) < +\infty \Rightarrow \nu(A_k) < \frac{1}{\min\limits_{y \in A_k} \{|y|^2, 1\}} \cdot (+\infty).}$$

Let $A_k = \{y : \frac{1}{k} \le |y|^2 \le k\}$. $\mathbb{R}^d \setminus \{0\} = \bigcup_{k=1}^{\infty} A_k$.

$\Rightarrow \nu(A_k) < k \int_{A_k} \min\{|y|^2, 1\} \nu(dy) < +\infty$. $\square$.

3. (a) <u>Proof</u>: Since ν is a Borel measure on $\mathbb{R}^d \setminus \{0\}$,

and the mapping $x \mapsto -x$ is continuous, mapping a Borel set in $\mathbb{R}^d \setminus \{0\}$ to a symmetric Borel set in $\mathbb{R}^d \setminus \{0\}$.

$\tilde{\nu}(A) = \nu(-A)$ is also a Borel measure.

ν is a Lévy measure $\Rightarrow \int_{\mathbb{R}^d \setminus \{0\}} \min\{|y|^2, 1\} \nu(dy) < +\infty$.

$$\Rightarrow \int_{\mathbb{R}^d\setminus\{0\}} \min\{|-y|^2, 1\}\ \nu(d(-y)) < +\infty.$$

$$\Rightarrow \int_{\mathbb{R}^d\setminus\{0\}} \min\{|y|^2, 1\}\ \tilde{\nu}(dy) < +\infty. \quad \Rightarrow \tilde{\nu} \text{ is a Lévy measure.} \ \square$$

(b) <u>Proof</u>: ① $-X_0 = 0$ a.s since $X_0 = 0$ a.s.

② $(-X_t) - (-X_s) = -(X_t - X_s)$.

X has indep. and stationary increments $X_t - X_s$

$\Rightarrow -X$ also has.

③ $\forall \varepsilon > 0$, $\lim_{t\to 0^+} \mathbb{P}(|X_t| > \varepsilon) = 0$

$\Rightarrow \forall \varepsilon > 0$, $\lim_{t\to 0^+} \mathbb{P}(|-X_t| > \varepsilon) = 0$.

$\Rightarrow -X$ is a Lévy process.

$$\Phi_{X_t}(u) = \exp\left\{t\left[ia^T u - \frac{1}{2}u^T A u + \int_{\mathbb{R}^d\setminus\{0\}} (e^{iu^T y} - 1 - iu^T y \mathbb{1}_{B_1(0)})\, \nu(dy)\right]\right\}.$$

$$\Phi_{-X_t}(u) = \mathbb{E}[e^{iu^T(-X_t)}] = \mathbb{E}[e^{-iu^T X_t}] = \mathbb{E}[e^{i(-u)^T X_t}] = \Phi_{X_t}(-u)$$

$$= \exp\left\{t\left[i(-a)^T u - \frac{1}{2}u^T A u + \int_{\mathbb{R}^d\setminus\{0\}} (e^{iu^T(-y)} - 1 - iu^T(-y)\mathbb{1}_{B_1(0)})\, \nu(dy)\right]\right\}$$

$$= \exp\left\{t\left[i(-a)^T u - \frac{1}{2}u^T A u + \int_{\mathbb{R}^d\setminus\{0\}} (e^{iu^T y} - 1 - iu^T y \mathbb{1}_{B_1(0)})\, \tilde{\nu}(dy)\right]\right\}.$$

$\Rightarrow (-a, A, \tilde{\nu})$ is the char. of $-X$. $\square$

4. <u>Sol</u>: A standard Brownian motion is a Lévy process by def.

Let its char by (a, A, ν).

$$\eta(x) = ia^T x - \frac{1}{2}x^T A x + \int_{\mathbb{R}^d\setminus\{0\}} (e^{ix^T y} - 1 - ix^T y \mathbb{1}_{B_1(0)}(y))\, \nu(dy).$$

$$\Phi_{B_t}(x) = e^{t\eta(x)}. \Rightarrow \Phi_{B_t - B_s}(x) = e^{(t-s)\eta(x)}.$$

Since $B_t - B_s \sim \mathcal{N}(\vec{0}_{d_1}, (t-s) I_{d_1})$, $\Phi_{B_t - B_s}(x) = e^{(t-s)(-\frac{1}{2}x^T x)}$.

Comparing above, one has: $a = \vec{0}_{d_1}$, $A = I_{d_1}$, $\nu = 0$.

$\Rightarrow$ Lévy symbol of B is $\eta(x) = -\frac{1}{2}x^T x$.

$$\Phi_{X_t}(x) = \mathbb{E}[e^{iu^T X_t}] = \mathbb{E}[e^{iu^T(at + PB_t)}]$$

$X_t = at + PB_t$.

① $X_0 = 0 + PB_0 = 0$ a.s. since $B_0 = 0$ a.s.

② $X_t - X_s = a(t-s) + P(B_t - B_s)$ only depends on $(t-s)$. $\Rightarrow$ stationary.

Also, $X_{t_1} - X_0, \cdots, X_{t_n} - X_{t_{n-1}}$ are indep.

③ $\forall \varepsilon > 0$, $\lim_{t \to 0^+} \mathbb{P}(|X_t| > \varepsilon) \leq \lim_{t \to 0^+} [\mathbb{P}(|at| > \varepsilon) + \mathbb{P}(|PX_t| > \varepsilon)] = 0$.

$\Rightarrow X$ is a Lévy process.

$$\Phi_{X_t}(x) = \mathbb{E}[e^{iu^T(at + PB_t)}] = e^{itu^T a} \cdot \mathbb{E}[e^{iu^T P B_t}]$$

$$= e^{itu^T a} \cdot \mathbb{E}[e^{i(P^T u)^T B_t}] = e^{itu^T a} \cdot e^{t(-\frac{1}{2}u^T P P^T u)}.$$

$$= e^{t(iu^T a - \frac{1}{2}u^T (PP^T) u)}.$$

$\Rightarrow (a, PP^T, 0)$ is the char. of X.

5. (a) <u>Proof</u>: Let the intensity of the Poisson process be $\lambda > 0$: $N = \{N_t : t \geq 0\}$.

$N_t \sim \text{Poisson}(\lambda t)$.

① A Poisson process is a 1-dim Lévy process by def.

② Since there exists a modification of N which is a RCLL Lévy process (here, again, a Poisson process), we may choose a RCLL version of N.

Since $N_t \sim \text{Poisson}(\lambda t)$,

$N_t - N_s \overset{d}{=} N_{t-s} \sim \text{Poisson}(\lambda(t-s))$, $\forall\, 0 \le s < t$.

$\Rightarrow N_t - N_s \in \{0, 1, 2, \cdots\}$ a.s.

$\Rightarrow \mathbb{P}(N_t \ge N_s) = 1$, $\forall\, 0 \le s < t$.

$\Rightarrow \bigcap_{\substack{p,q \in \mathbb{Q} \\ 0 \le p < q}}^{\infty} \mathbb{P}(N_p \le N_q) = 1$. i.e. $\mathbb{P}(N_p \le N_q\ \forall\, 0 \le p < q,\ p,q \in \mathbb{Q}) = 1$.

Since N_t is RCLL and takes finite value,

$\mathbb{P}(N_t \le N_s\ \forall\, 0 \le t < s) = 1$. i.e. N is increasing a.s. $\square$.

(b) <u>Proof</u>: $Y = \{Y_t : t \ge 0\}$. $Y_t = \sum_{n=1}^{N_t} Z_n$, Z_n: i.i.d. $N = \{N_t : t \ge 0\}$ – Poisson process.

① We've proved in class that a compound Poisson process is a Lévy process.

② $\forall\, 0 \le s < t$, $Y_t - Y_s = \sum_{n=N_s+1}^{N_t} Z_n$

By (a), $N_t \ge N_s$ a.s. Also, $Z_n \ge 0$ a.s. $\forall n$. $\Rightarrow Y_t - Y_s \ge 0$ a.s.

$\Rightarrow Y$ is increasing a.s. $\square$.

6. <u>Proof</u>: The problem itself implies X is RCLL. $X_t = \sum_{n=1}^{N_t} Z_n$.

$$\forall\, t > 0,\ \sum_{0 \le s \le t} |\Delta X_s| = \sum_{0 \le s \le t} |X_s - X_{s-}| = \sum_{0 \le s \le t} \left| \sum_{n=1}^{N_s} Z_n - \sum_{n=1}^{N_{s-}} Z_n \right|$$

If $N_s = N_{s-}$, $\Delta X_s = 0$.

If $N_s \ne N_{s-}$, $N_s = N_{s-} + 1$.

$$= \sum_{\substack{0 \le s \le t \\ N_s \ne N_{s-}}} |Z_{N_s}| = \sum_{n=1}^{N_t} |Z_n|.$$

Since $N_t < +\infty$ a.s. $\sum_{0 \le s \le t} |\Delta X_s| = \sum_{n=1}^{N_t} |Z_n| < \infty$ a.s. $\square$.

$N_t \sim \text{Poisson}(\lambda t)$

$\Rightarrow \mathbb{P}(N_t < +\infty) = 1$. t: finite.